Self Consistency

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Self-Consistency Improves Chain of Thought Reasoning in Language Models Authors: Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, Denny Zhou Published: ICLR 2023

Abstract: We explore a simple ensemble strategy, self-consistency, that significantly improves the reasoning accuracy of large language models. The approach involves sampling a diverse set of reasoning paths instead of only taking the greedy one, and then selecting the most consistent answer by marginalizing out the sampled reasoning paths. Self-consistency leverages the intuition that a complex reasoning problem typically admits multiple different ways of thinking leading to its unique correct answer.

Core Method: Self-Consistency is a decoding strategy that improves chain-of-thought reasoning by sampling multiple reasoning paths and taking a majority vote on the final answer.

The Self-Consistency Process: 1. Prompt with Chain-of-Thought: - Use standard CoT prompting with examples - Instead of greedy decoding, use sampling (temperature > 0) - Generate multiple diverse reasoning paths

2. Sample Diverse Reasoning Paths: - Generate k different completions for the same problem - Each completion may use different reasoning steps - Sampling introduces diversity in approaches - Typical k values: 5, 10, 20, 40

3. Extract Final Answers: - Parse the final answer from each reasoning path - Answers may be numerical, multiple choice, or short text - Handle formatting variations in answers

4. Majority Vote: - Count the frequency of each answer - Select the most common answer as the final result - This is equivalent to marginalizing over reasoning paths - The intuition: correct answers are more likely to be reached consistently

Key Features: - No additional training required - purely a decoding strategy - Works with any CoT prompt - Improves accuracy without changing the model - Provides robustness against reasoning errors - The diversity of paths naturally handles different solution approaches

Why Self-Consistency Works: - Correct reasoning paths tend to converge on the same answer - Incorrect paths may produce different wrong answers - Majority voting cancels out random errors - The approach leverages the "wisdom of crowds" within one model - Sampling explores alternative solution methods

Experiments: Self-Consistency was evaluated on various reasoning benchmarks: - GSM8K: Improved from 57.2% to 74.4% (with k=40) - SVAMP: Improved from 79.1% to 86.6% - AQuA: Improved from 34.5% to 44.2% - StrategyQA: Improved from 70.2% to 75.2% - ARC-Challenge: Improved from 85.2% to 87.5% - Consistent improvements across all benchmarks

Key Findings: - Self-consistency significantly improves CoT reasoning - The improvement increases with more samples (diminishing returns) - The method is most effective for complex multi-step problems - It works across different model sizes and architectures - Self-consistency can reveal uncertainty - if answers are split, the model is unsure - The approach is simple and cost-effective compared to fine-tuning